DESCENT OF FAMILIES OF R-CYCLES ALONG MORPHISMS WITH SEPARABLE RESIDUE FIELD EXTENSIONS

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Allowed standard prerequisites (editorial).

- Injectivity of base change for cycles over fields: for a locally algebraic scheme X over a field k, the base change map Z_r(X) to Z_r(X_{k'}) on r-cycles is always injective (relative-cycles chapter, Section section-relative-fields, canonical tag 0H4E). cited at proof lines 848-850. Yields part (1): the base change map on families of r-cycles over X/S is injective whenever S' to S is surjective.
- Positivity of multiplicities in field extensions: the multiplicity m_{Z', Z_K} of an irreducible component Z' of the base change of an integral locally algebraic k-scheme Z by a field extension K/k is one or a power of the characteristic of k, in particular a positive integer (relative-cycles chapter, Lemma lemma-multiplicities-field-extension, canonical tag 0H4F). cited at proof lines 880-881. The coefficients m(x', g) in the base change operation on coefficient functions are positive integers.
- Existence and coequalizer property of lifts of points of transdegree r under a surjective base change: every x of transdegree r admits a lift x' of transdegree r, and any two such lifts are both projections of a common x" of transdegree r in the double base change (relative-cycles chapter, Lemma lemma-coequalizer-dim-r, canonical tag 0H4W). cited at proof lines 899-903. Together with the displayed multiplicity identity and the footnote construction, gives a unique rational-valued function alpha with g^*alpha = alpha'.

Source reference. Original section 0H4E: sources/stacks-chow/tags.txt, source lines 21349–21349. Cited at proof lines 848-850: part (1) rests on the injectivity of the base change map established there.

Source reference. Original lemma 0H4F: sources/stacks-chow/tags.txt, source lines 21350–21350. Cited at proof lines 880-881: source of the positivity of the coefficients m(x', g) in the base change operation on coefficient functions.

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0H4W: sources/stacks-chow/tags.txt, source lines 21366–21366. Cited at proof lines 899-903: with the footnote construction yields the unique rational-valued function alpha with g^*alpha = alpha'.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $r \geq 0$ be an integer. Let $g : S' \to S$ be a morphism of schemes and $X' = S' \times_S X$. Assume that for every $s \in S$ there exists a point $s' \in S'$ with $g(s') = s$ and such that $\kappa(s')/\kappa(s)$ is a separable extension of fields. Then*

1. *For families α_1 and α_2 of r-cycles on fibres of X/S if $g^*\alpha_1 = g^*\alpha_2$, then $\alpha_1 = \alpha_2$.*
2. *Given a family α' of r-cycles on fibres of X'/S' if $pr_1^*\alpha' = pr_2^*\alpha'$ as families of r-cycles on fibres of $(S' \times_S S') \times_S X/(S' \times_S S')$, then there is a unique family α of r-cycles on fibres of X/S such that $g^*\alpha = \alpha'$.*

Proof. Part (1) follows from the injectivity of the base change map discussed in Section 0H4E. (This argument works as long as $S' \to S$ is surjective.)

Let α' be as in (2). Denote $\alpha'' = \mathrm{pr}_1^*\alpha' = \mathrm{pr}_2^*\alpha'$ the common value.

Let $(X/S)^{(r)}$ be the set of $x \in X$ with $\mathrm{trdeg}_{\kappa(f(x))}(\kappa(x)) = r$ and similarly define $(X'/S')^{(r)}$ and $(X''/S'')^{(r)}$ Taking coefficients, we may think of α' and α'' as functions $\alpha' : (X'/S')^{(r)} \to \mathbf{Z}$ and $\alpha'' : (X''/S'')^{(r)} \to \mathbf{Z}$. Given a function

$$\varphi : (X/S)^{(r)} \to \mathbf{Z}$$

we define $g^*\varphi : (X'/S')^{(r)} \to \mathbf{Z}$ by analogy with our base change operation. Namely, say $x' \in (X'/S')^{(r)}$ maps to $x \in X$, $s' \in S'$, and $s \in Z$. Denote $Z' \subset X'_{s'}$ and $Z \subset X_s$ the integral closed subschemes with generic points x' and x. Note that $\dim(Z') = r$. If $\dim(Z) < r$, then we set $(g^*\varphi)(x') = 0$. If $\dim(Z) = r$, then Z' is an irreducible component of $Z_{s'}$ and hence has a multiplicity $m_{Z',Z_{s'}}$. Call this $m(x', g)$. Then we define

$$(g^*\varphi)(x') = m(x', g)\varphi(x)$$

Note that the coefficients $m(x', g)$ are always positive integers (see for example Lemma 0H4F). We similarly have base change maps

$$\mathrm{pr}_1^*, \mathrm{pr}_2^* : \mathrm{Map}((X'/S')^{(r)}, \mathbf{Z}) \longrightarrow \mathrm{Map}((X''/S'')^{(r)}, \mathbf{Z})$$

It follows from the associativity of base change that we have $\mathrm{pr}_1^* \circ g^* = \mathrm{pr}_2^* \circ g^*$ (small detail omitted). To be explicitly, in terms of the maps of sets this equality just means that for $x'' \in (X''/S'')^{(r)}$ we have

$$m(x'', \mathrm{pr}_1)m(\mathrm{pr}_1(x''), g) = m(x'', \mathrm{pr}_2)m(\mathrm{pr}_2(x''), g)$$

provided that $\mathrm{pr}_1(x'')$ and $\mathrm{pr}_2(x'')$ are in $(X''/S'')^{(r)}$. By Lemma 0H4W and an elementary argument[1] using the previous displayed equation, it follows that there exists a unique map

$$\alpha : (X/S)^{(r)} \to \mathbf{Q}$$

such that $g^*\alpha = \alpha'$. To finish the proof it suffices to show that α has integer values (small detail omitted: one needs to see that α determines a locally finite sum on each fibre which follows from the corresponding fact for α'). Given any

[1] Given $x \in (X/S)^{(r)}$ pick $x' \in (X'/S')^{(r)}$ mapping to x and set $\alpha(x) = \alpha'(x')/m(x', g)$. This is well defined by the formula and the lemma.

$x \in (X/S)^{(r)}$ with image $s \in S$ we can pick a point $s' \in S'$ such that $\kappa(s')/\kappa(s)$ is separable. Then we may choose $x' \in (X'/S')^{(r)}$ mapping to s and x and we see that $m(x', g) = 1$ because $Z_{s'}$ is reduced in this case. Whence $\alpha(x) = \alpha'(x')$ is an integer. □